Feedforward Networks

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```
# The Python chunks on this page import these; reticulate provisions them.
# Attaching reticulate gives the inline R its `py` object.
library(reticulate)
source("../tools/use-torch.R")
use_torch(c("numpy", "pandas", "matplotlib"))
```

Brian Hutchinson's CSCI 481/581 sets the order of topics in these notes, which follow it with his permission; this page covers the layered networks of his module on machine learning[1] fundamentals. The exercises are new. The output units and universal approximation follow *Deep Learning*, Chapter 6 [1].

1 Welcome

1.1 Plan for this lecture

The PyTorch chapter fit the soft 0-or-1 rule, a model with one parameter[2]. This lecture builds models that learn their own features[3] of the digits:

1. what a stack of linear layers can and cannot compute, and what a function applied between the layers changes;
2. how a network turns its last layer's numbers into class probabilities and a loss, and how PyTorch computes that loss without overflowing;
3. how much a network with one layer of learned features[4] can represent.

1.2 The digits as tensors

```
import numpy as np
import pandas as pd
import torch
from torch import nn
import matplotlib.pyplot as plt

_ = torch.manual_seed(481)
digits = pd.read_csv("data/optdigits-test.csv", header=None)
x = torch.tensor(digits.iloc[:, :64].to_numpy(), dtype=torch.float32) / 16
y = torch.tensor(digits[64].to_numpy())
n_train = 1200
order = torch.randperm(len(x))
x_train, y_train = x[order[:n_train]], y[order[:n_train]]
x_test, y_test = x[order[n_train:]], y[order[n_train:]]
n_test = len(x_test)
print(x_train.shape, x_test.shape)
```

[1] https://morrison-lab.github.io/lds/chapters/course-overview.html#def-machine-learning
[2] https://morrison-lab.github.io/lds/chapters/big-ideas.html#def-parameter
[3] https://morrison-lab.github.io/lds/chapters/anatomy.html#def-feature
[4] https://morrison-lab.github.io/lds/chapters/anatomy.html#def-feature

```
torch.Size([1200, 64]) torch.Size([597, 64])
```

The exercises use the 1,797 handwritten digits from the course overview, from the same local copy, `data/optdigits-test.csv`[5], of the UCI Machine Learning Repository's data [2]. Each image's 64 pixels, from 0 to 16, are divided by 16 so that they run from 0 to 1, and a random 1,200 images form a training set[6], with the other images as a test set[7].

2 Layers

2.1 Two layers with nothing between them

Exercise 2.1 (Two layers with nothing between them)**.** A layer `first = nn.Linear(64, 32)` maps a digit's 64 pixels $\underset{\sim}{x}$ to 32 numbers $\mathbf{W}_1\underset{\sim}{x} + \underset{\sim}{b}_1$, and a layer `second = nn.Linear(32, 10)` maps 32 numbers $\underset{\sim}{u}$ to 10 numbers $\mathbf{W}_2\underset{\sim}{u} + \underset{\sim}{b}_2$. What kind of function of $\underset{\sim}{x}$ does the stack `second(first(x))` compute?

Solution

Solution 2.1. Substituting the first layer into the second,

$$\begin{aligned}\mathbf{W}_2(\mathbf{W}_1\underset{\sim}{x} + \underset{\sim}{b}_1) + \underset{\sim}{b}_2 &= \mathbf{W}_2\mathbf{W}_1\underset{\sim}{x} + \mathbf{W}_2\underset{\sim}{b}_1 + \underset{\sim}{b}_2 && \text{(distribute } \mathbf{W}_2\text{)}\\ &= \underbrace{(\mathbf{W}_2\mathbf{W}_1)}_{10\times 64}\underset{\sim}{x} + \underbrace{(\mathbf{W}_2\underset{\sim}{b}_1 + \underset{\sim}{b}_2)}_{10\times 1} && \text{(group the terms that do not involve } \underset{\sim}{x}\text{)}\end{aligned}$$

So the stack is an affine function[a] of $\underset{\sim}{x}$, the same kind of function as one `nn.Linear(64, 10)`, with weight matrix $\mathbf{W}_2\mathbf{W}_1$ and bias $\mathbf{W}_2\underset{\sim}{b}_1 + \underset{\sim}{b}_2$. The 32 numbers in the middle add nothing that one layer could not compute.

```
first = nn.Linear(64, 32)
second = nn.Linear(32, 10)
weight = second.weight @ first.weight
bias = second.weight @ first.bias + second.bias
stacked = second(first(x_test))
one_layer = x_test @ weight.T + bias
stack_gap = (stacked - one_layer).abs().max().item()
print(weight.shape, bias.shape, stack_gap)
```

```
torch.Size([10, 64]) torch.Size([10]) 1.4901161193847656e-07
```

On the 597 test images, the stack and the single layer agree to within 1.5e-07, a rounding error. `nn.Linear` stores $\mathbf{W}$ with one row per output, and computes `x @ weight.T + bias` for a tensor `x` with one row per image.

[a] https://morrison-lab.github.io/lds/chapters/linear-regression.html#def-affine-function

2.2 A function between the layers

Exercise 2.2 (A function between the layers)**.** Suppose the logistic function[a], $\text{expit}(z) = 1/(1+\mathrm{e}^{-z})$, is applied to each of the 32 numbers between the two layers of Exercise 2.1, giving `second(torch.sigmoid(first(x)))`. Is that stack still an affine function[b] of $\underset{\sim}{x}$?

Hint

An affine function[c] f maps the midpoint of two inputs to the midpoint of their outputs: $f\left(\frac{1}{2}(\underset{\sim}{x} + \underset{\sim}{x}')\right) = \frac{1}{2}(f(\underset{\sim}{x}) + f(\underset{\sim}{x}'))$. Try two digits.

[a] https://morrison-lab.github.io/mds/functions.html#def-logistic-function

[5] https://morrison-lab.github.io/dds/chapters/data/optdigits-test.csv
[6] https://morrison-lab.github.io/lds/chapters/big-ideas.html#def-training-set
[7] https://morrison-lab.github.io/lds/chapters/big-ideas.html#def-test-set

[b] https://morrison-lab.github.io/lds/chapters/linear-regression.html#def-affine-function
[c] https://morrison-lab.github.io/lds/chapters/linear-regression.html#def-affine-function

Solution

Solution 2.2.

```
def midpoint_gap(f, a, b):
    # Largest gap between f at the midpoint and the midpoint of f.
    return (f((a + b) / 2) - (f(a) + f(b)) / 2).abs().max().item()

a, b = x_test[0], x_test[1]
gap_linear = midpoint_gap(lambda v: second(first(v)), a, b)
gap_logistic = midpoint_gap(lambda v: second(torch.sigmoid(first(v))), a, b)
mean_abs_pre = first(torch.stack([a, b])).abs().mean().item()
print(gap_linear, gap_logistic, mean_abs_pre)
```

```
5.960464477539063e-08 0.0003224015235900879 0.2633136212825775
```

No. Take the first two test images, which show the digits 2 and 8. For them, the stack with nothing between its layers maps their midpoint to the midpoint of their outputs, to within 6e-08, a rounding error. With expit between the layers it misses by up to 0.00032, far more than a rounding error, so it is not an affine function[a]. Even so, expit bends the untrained stack only a little. The untrained first layer's numbers for these two images have a mean absolute value of 0.26, and near 0, expit is close to its tangent line, $\frac{1}{2} + \frac{z}{4}$. Training can move the numbers to where expit bends.
The logistic function[b] bends each of the 32 numbers, so each of the 10 outputs is an affine function of 32 bent features[c], not of the pixels. Training moves $\mathbf{W}_1$ and $\underset{\sim}{b}_1$ as well as $\mathbf{W}_2$ and $\underset{\sim}{b}_2$, so it chooses those features.

[a] https://morrison-lab.github.io/lds/chapters/linear-regression.html#def-affine-function
[b] https://morrison-lab.github.io/mds/functions.html#def-logistic-function
[c] https://morrison-lab.github.io/lds/chapters/anatomy.html#def-feature

Definition 2.1 (Hidden layer)**.** A **hidden layer** of a network is a vector that the network computes from its input on the way to its output, and that is neither the input nor the output [1, Ch. 6].
The training data[a] give an input and a target output for each example, but no target for a hidden layer: training chooses the parameters[b] that compute it. These notes write a hidden layer as $\underset{\sim}{h}$.

[a] https://morrison-lab.github.io/lds/chapters/anatomy.html#def-training-data
[b] https://morrison-lab.github.io/lds/chapters/big-ideas.html#def-parameter

Example 2.1 (The 32 hidden values of one digit)**.**

```
hidden = torch.sigmoid(first(x_test[0]))
print(hidden.shape)
```

```
torch.Size([32])
```

```
print(hidden.detach().numpy().round(2))
```

```
[0.39 0.42 0.48 0.41 0.57 0.49 0.51 0.51 0.54 0.6  0.44 0.53 0.64 0.41
 0.44 0.63 0.61 0.41 0.43 0.56 0.49 0.54 0.56 0.56 0.46 0.42 0.59 0.46
 0.55 0.54 0.6  0.46]
```

In Solution 2.2, `torch.sigmoid(first(x))` is a hidden layer of width[a] 32. For the first test image it holds the 32 numbers printed, each between 0 and 1. `.detach()` drops the record that automatic differentiation keeps, so that NumPy can print the values.

[a] https://morrison-lab.github.io/lds/chapters/neural-nets.html#def-width

Definition 2.2 (Unit (neuron))**.** A **unit**, or **neuron**, is one entry of a hidden layer [1, Ch. 6]. The number of units in a hidden layer is its width[a].

[a] https://morrison-lab.github.io/lds/chapters/neural-nets.html#def-width

Example 2.2 (One unit of the digits network)**.** The hidden layer of Example 2.1 has 32 units. For the first test image, unit 1 holds 0.39, the first number printed there.

Definition 2.3 (Activation function (nonlinearity))**.** An **activation function**, or **nonlinearity**, is a fixed function $g : \mathbb{R} \to \mathbb{R}$, not an affine function[a], that a network applies to each entry of an affine function of the layer before, so that a hidden layer is

$$\underset{\sim}{h} = g(\mathbf{W}\underset{\sim}{x} + \underset{\sim}{b}),$$

with g applied entry by entry [1, Sec. 6.3].

[a] https://morrison-lab.github.io/lds/chapters/linear-regression.html#def-affine-function

Example 2.3 (The logistic function as an activation function)**.**

```
pre = first(x_test[0])
print(pre[:4].detach().numpy().round(3))
```

```
[-0.458 -0.31  -0.068 -0.356]
```

```
print(torch.sigmoid(pre[:4]).detach().numpy().round(3))
```

```
[0.387 0.423 0.483 0.412]
```

In Example 2.1 the activation function is the logistic function[a], $g(z) = \text{expit}(z) = 1/(1 + \mathrm{e}^{-z})$. The first line printed holds the first 4 entries of $\mathbf{W}_1\underset{\sim}{x} + \underset{\sim}{b}_1$ for the first test image, and the second holds the logistic function of each. The logistic function maps each entry separately: entry 1 of the output depends only on entry 1 of the input.

[a] https://morrison-lab.github.io/mds/functions.html#def-logistic-function

Definition 2.4 (Pre-activation)**.** The **pre-activation** of a hidden layer $\underset{\sim}{h} = g(\mathbf{W}\underset{\sim}{x} + \underset{\sim}{b})$ is the vector $\mathbf{W}\underset{\sim}{x} + \underset{\sim}{b}$ that goes into its activation function g.

Example 2.4 (The pre-activation of one digit)**.** In Example 2.3, the first line printed holds the first 4 entries of the pre-activation of the hidden layer of Example 2.1, for the first test image, and the second line holds the same entries of the hidden layer itself.

2.3 Three activation functions

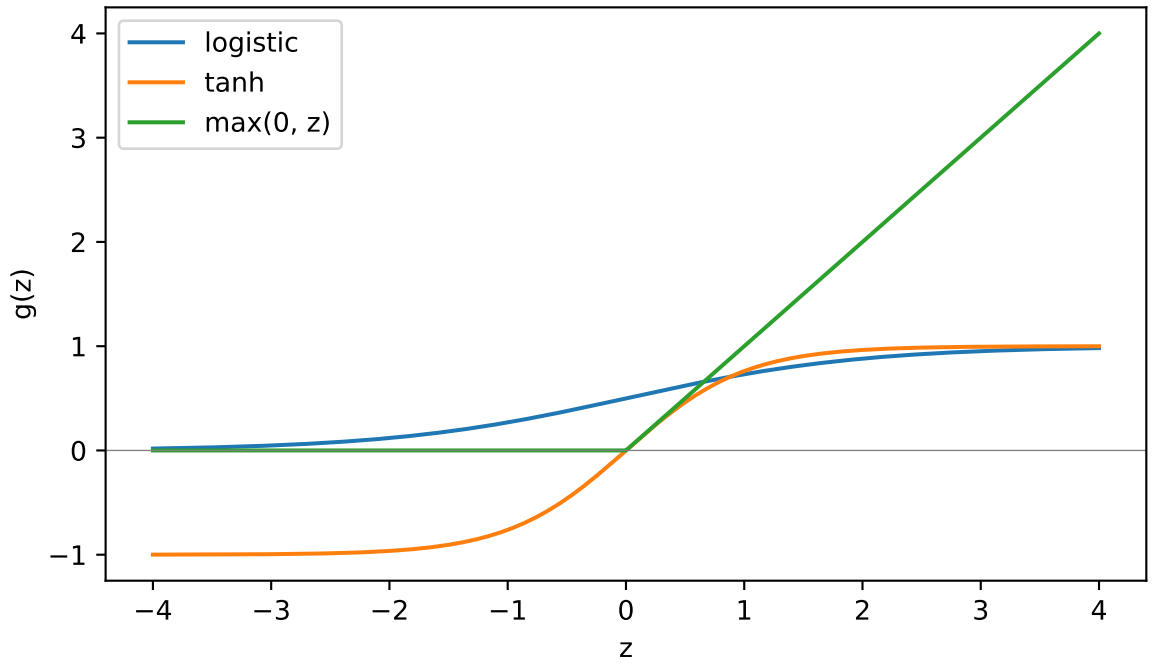
Exercise 2.3 (Three activation functions)**.** How do the logistic function[a] $\text{expit}(z)$, the hyperbolic tangent $\tanh(z)$, and the function $\max(0, z)$ behave as $|z|$ grows large?

[a] https://morrison-lab.github.io/mds/functions.html#def-logistic-function

Solution

Solution 2.3.

```
z = torch.linspace(-4, 4, 401)
activations = {
    "logistic": torch.sigmoid(z),
    "tanh": torch.tanh(z),
    "max(0, z)": torch.relu(z),
}
fig, ax = plt.subplots(figsize=(6, 3.5), layout="constrained")
for name, value in activations.items():
    ax.plot(z, value, label=name);
ax.axhline(0, color="gray", linewidth=0.5);
ax.set_xlabel("z");
ax.set_ylabel("g(z)");
ax.legend();
plt.show()
```



- **Large positive** z**.** expit(z) levels off at 1, and $\tanh(z)$ levels off at 1, while $\max(0, z) = z$ keeps growing.
- **Large negative** z**.** expit(z) levels off at 0, and $\tanh(z)$ levels off at -1, while $\max(0, z)$ is exactly 0 for every negative z.

tanh is a rescaled logistic function[a], $\tanh(z) = 2\text{expit}(2z) - 1$, so its graph is the same S curve, centred on 0 instead of $\frac{1}{2}$.

[a] https://morrison-lab.github.io/mds/functions.html#def-logistic-function

Definition 2.5 (Rectified linear unit (ReLU))**.** The **rectified linear unit**, or **ReLU**, is the activation function

$$\text{ReLU}(z) = \max(0, z)$$

[1, Sec. 6.3.1], [3]. PyTorch provides it as `torch.relu` and as the layer `nn.ReLU()`. In mathematics $\max(0, z)$ is also called the positive part of z, written z^+. *Deep Learning* recommends ReLU as the default activation function for hidden layers [1, Sec. 6.1].

Example 2.5 (ReLU on the digits)**.**

```
hidden_relu = torch.relu(first(x_test))
zero_share = (hidden_relu == 0).float().mean().item()
print(hidden_relu.shape, zero_share)
```

```
torch.Size([597, 32]) 0.4586997628211975
```

Applying the ReLU to the pre-activations of the first layer of Exercise 2.1, for all 597 test images, gives a hidden layer of width[a] 32 for each image. 46% of its entries are exactly 0, one for each negative pre-activation.

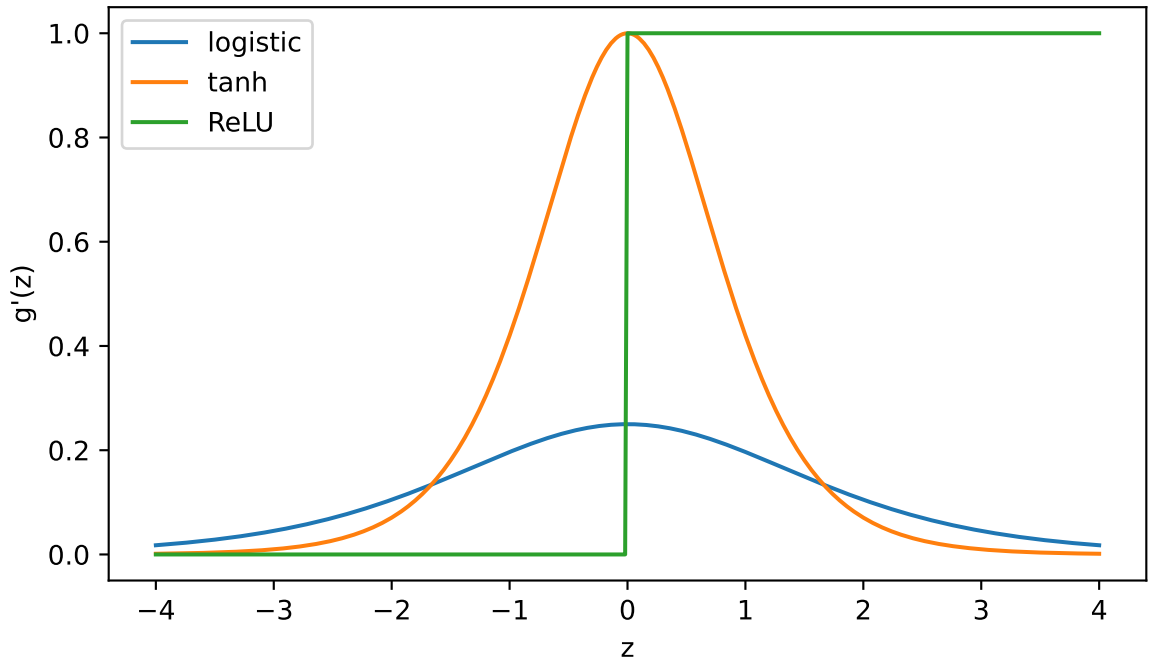
[a] https://morrison-lab.github.io/lds/chapters/neural-nets.html#def-width

Exercise 2.4 (Where the slopes vanish)**.** Using automatic differentiation, find the values of z at which the derivative of each activation function of Exercise 2.3 is close to 0.

Solution

Solution 2.4.

```
z = torch.linspace(-4, 4, 401, requires_grad=True)
fig, ax = plt.subplots(figsize=(6, 3.5), layout="constrained")
slopes = {}
for name, g in [("logistic", torch.sigmoid), ("tanh", torch.tanh), ("ReLU", torch.relu)]:
    (slope,) = torch.autograd.grad(g(z).sum(), z)
    slopes[name] = slope
    ax.plot(z.detach(), slope, label=name);
ax.set_xlabel("z");
ax.set_ylabel("g'(z)");
ax.legend();
plt.show()
```




```
largest_logistic_slope = slopes["logistic"].max().item()
logistic_slope_at_4 = slopes["logistic"][-1].item()
```

Each output of `g(z)` depends only on the matching entry of `z`, so the gradient of `g(z).sum()` holds the

derivative $g'(z)$ at each point.

- **Logistic.** Its derivative peaks at 0.25 at $z = 0$, and is close to 0 once $|z|$ is large: 0.018 at $z = 4$.
- **tanh.** Its derivative peaks at 1 at $z = 0$ and is also close to 0 for large $|z|$.
- **ReLU.** Its derivative is 1 for every positive z and exactly 0 for every negative z. At $z = 0$ it has no derivative; PyTorch uses 0 there.

A unit whose pre-activation sits where the derivative is near 0 is saturated[a]: a small change in its weights barely changes its output, so the gradient that training follows is small too. The logistic function[b] and tanh saturate on both sides, and ReLU only on its negative side.

[a] https://morrison-lab.github.io/lds/chapters/backpropagation.html#def-saturation
[b] https://morrison-lab.github.io/mds/functions.html#def-logistic-function

2.4 Feedforward networks

Definition 2.6 (Feedforward network (multilayer perceptron, MLP))**.** A **feedforward network**, or **multilayer perceptron** (**MLP**), computes its output $\underset{\sim}{\hat{y}}$ from its input $\underset{\sim}{x}$ through a sequence of K hidden layers,

$$\begin{aligned} \underset{\sim}{h}^{(1)} &= g\big(\mathbf{W}^{(1)}\underset{\sim}{x} + \underset{\sim}{b}^{(1)}\big), \\ \underset{\sim}{h}^{(\ell)} &= g\big(\mathbf{W}^{(\ell)}\underset{\sim}{h}^{(\ell-1)} + \underset{\sim}{b}^{(\ell)}\big), \qquad \ell = 2, \dots, K, \\ \underset{\sim}{\hat{y}} &= \mathbf{W}^{(K+1)}\underset{\sim}{h}^{(K)} + \underset{\sim}{b}^{(K+1)}, \end{aligned}$$

where g is an activation function and the weight matrices $\mathbf{W}^{(\ell)}$ and bias vectors $\underset{\sim}{b}^{(\ell)}$ are the network's parameters[a] [1, Ch. 6]. Information flows from the input to the output, with no path from a later layer back to an earlier one; "feedforward" names that.

Every layer of an MLP is fully connected[b], so it is also called a fully connected network, and *Deep Learning* calls it a deep feedforward network. It contains no perceptron in Rosenblatt's sense, a threshold function of a weighted sum [4], so the name is historical. Its depth[c] is $K + 1$. The last layer has no activation function; how its numbers become predictions depends on the task.

[a] https://morrison-lab.github.io/lds/chapters/big-ideas.html#def-parameter
[b] https://morrison-lab.github.io/lds/chapters/neural-nets.html#def-fully-connected
[c] https://morrison-lab.github.io/lds/chapters/neural-nets.html#def-depth

Example 2.6 (An MLP for the digits)**.**

```
_ = torch.manual_seed(1)
mlp = nn.Sequential(
    nn.Linear(64, 32),
    nn.ReLU(),
    nn.Linear(32, 10),
)
print(mlp)
```

```
Sequential(
  (0): Linear(in_features=64, out_features=32, bias=True)
  (1): ReLU()
  (2): Linear(in_features=32, out_features=10, bias=True)
)
```

```
scores = mlp(x_test[:5])
print(scores.shape)
```

```
torch.Size([5, 10])
```

`nn.Sequential` applies its layers in order, so `mlp` is a feedforward network with one hidden layer of width[a] 32, the ReLU as its activation function, and 10 outputs, one per digit. For 5 test images it

returns a tensor of shape (5, 10): 10 numbers for each image.

[a] https://morrison-lab.github.io/lds/chapters/neural-nets.html#def-width

Remark (Writing a layer)

Remark 2.1 (Writing a layer). One layer is written here as $g(\mathbf{W}\underset{\sim}{x} + \underset{\sim}{b})$, with $\underset{\sim}{x}$ a column vector and $\mathbf{W}$ having one row per output. Other sources write it differently:

- PyTorch's `nn.Linear` stores $\mathbf{W}$ in the same orientation, but takes a batch of inputs with one row per example, and computes `x @ weight.T + bias`.
- *Deep Learning* writes $g(\mathbf{W}^\top \underset{\sim}{x} + \underset{\sim}{c})$, with one column of $\mathbf{W}$ per output [1, Sec. 6.1].
- lds[a] puts a 1 at the start of $\underset{\sim}{x}$ and folds the bias into $\mathbf{W}$.

[a] https://morrison-lab.github.io/lds/chapters/neural-nets.html#sec-learning-the-features

Read

Deep Learning's Chapter 6[a] covers feedforward networks, their output units and their activation functions [1, Ch. 6].

[a] https://www.deeplearningbook.org/contents/mlp.html

3 Output units and losses

3.1 From scores to probabilities

Exercise 3.1 (From scores to probabilities)**.** The MLP of Example 2.6 gives each image 10 real numbers, positive or negative. How can they become 10 probabilities, one per digit, each positive and adding up to 1?

Solution

Solution 3.1. Apply the softmax function[a]: exponentiate each number, which makes it positive, then divide by the sum of the exponentials, which makes them add up to 1.

```
probabilities = torch.softmax(scores, dim=1)
print(probabilities.detach().numpy().round(3))
```

```
[[0.094 0.092 0.106 0.102 0.141 0.098 0.083 0.104 0.099 0.081]
 [0.084 0.091 0.121 0.093 0.153 0.085 0.073 0.109 0.109 0.082]
 [0.094 0.084 0.121 0.097 0.142 0.089 0.082 0.098 0.113 0.08 ]
 [0.09  0.088 0.119 0.1   0.143 0.087 0.076 0.106 0.103 0.087]
 [0.09  0.088 0.129 0.1   0.144 0.084 0.075 0.108 0.105 0.075]]
```

```
print(probabilities.sum(dim=1))
```

```
tensor([1.0000, 1.0000, 1.0000, 1.0000, 1.0000], grad_fn=<SumBackward1>)
```

`dim=1` applies softmax across the 10 numbers in each row, one row per image. Each row of probabilities adds up to 1, and each is near $\frac{1}{10}$, since the network is untrained.

[a] https://morrison-lab.github.io/lds/chapters/multiclass-logistic-regression.html#def-softmax

Definition 3.1 (Logits)**.** The **logits** of a classification network are the numbers its output layer computes for each class, before the softmax function[a] turns them into probabilities. PyTorch's documentation uses this name.

The name comes from the logit function, the log odds[b], but most logits are not log odds. Adding one number to every logit leaves the softmax unchanged, so only their differences matter: with two classes, the difference $z_2 - z_1$ is the log odds of class 2. Other names for logits are scores and unnormalized log probabilities.

[a] https://morrison-lab.github.io/lds/chapters/multiclass-logistic-regression.html#def-softmax
[b] https://morrison-lab.github.io/pds/probability-basics.html#def-log-odds

Example 3.1 (Shifting the logits)**.**

```
logits = mlp(x_test[:1])
shifted = logits + 5
softmax_gap = (torch.softmax(shifted, dim=1) - torch.softmax(logits, dim=1)).abs().max().item()
print(logits.detach().numpy().round(3))
```

```
[[ 0.    -0.024  0.112  0.08   0.404  0.04  -0.133  0.095  0.045 -0.158]]
```

```
print(softmax_gap)
```

```
2.9802322387695312e-08
```

The logits of the first test image are the 10 numbers printed. Adding 5 to each changes the softmax probabilities by at most 3e-08, a rounding error.

Exercise 3.2 (The loss before training)**.** The untrained MLP of Example 2.6 gives each test image probabilities near $\frac{1}{10}$. About what mean cross-entropy[a] loss should it have on the test set[b]?

[a] https://morrison-lab.github.io/pds/expectation.html#def-cross-entropy
[b] https://morrison-lab.github.io/lds/chapters/big-ideas.html#def-test-set

Solution

Solution 3.2. The cross-entropy[a] loss of one image is $-\log$ of the probability the network gives its true class. If every probability is $\frac{1}{10}$, every image's loss is $-\log \frac{1}{10} = \log 10$, and so is their mean.

```
start_loss = nn.functional.cross_entropy(mlp(x_test), y_test).item()
print(start_loss, np.log(10))
```

```
2.3070168495178223 2.302585092994046
```

`nn.functional.cross_entropy` takes the logits and the true digits, and returns the mean loss over the test set[b], 2.307, close to $\log 10 \approx 2.303$. A training run whose first loss is far from $\log C$, for C classes, is worth checking for a bug before it runs further.

[a] https://morrison-lab.github.io/pds/expectation.html#def-cross-entropy
[b] https://morrison-lab.github.io/lds/chapters/big-ideas.html#def-test-set

3.2 Computing the loss from logits

Exercise 3.3 (Softmax straight from its formula)**.** A trained network can be confident, with logits in the hundreds. Take `z` to be the first test image's logits from Example 3.1 times 1,000. What goes wrong when the softmax of `z` is computed with its formula, `torch.exp(z) / torch.exp(z).sum()`?

Solution

Solution 3.3.

```
z = (1000 * logits[0]).detach()
naive = torch.exp(z) / torch.exp(z).sum()
largest_float = torch.finfo(torch.float32).max
print(z.numpy().round(1))
```

```
[ 4.000e-01 -2.380e+01  1.118e+02  7.990e+01  4.042e+02  3.990e+01
 -1.331e+02  9.520e+01  4.500e+01 -1.576e+02]
```

```
print(naive)
```

```
tensor([0., 0., nan, 0., nan, 0., 0., nan, 0., 0.])
```

```
print(np.log(largest_float))
```

```
88.72283905206835
```

```
n_nan = torch.isnan(naive).sum().item()
n_zero = (naive == 0).sum().item()
print(n_nan, n_zero)
```

```
3 7
```

The result holds 3 `nan`s and 7 zeros. A `float32` number cannot be larger than about 3.4×10^{38}, so `torch.exp(z)` is infinite for every z above the log of that number, about 88.7, and infinity divided by infinity is `nan`. Every other entry has a finite exponential divided by that infinite sum, so its probability is 0, and the log of 0 is $-\infty$.

Exercise 3.4 (The log of a softmax probability)**.** The cross-entropy[a] loss needs $\log \operatorname{softmax}(\underset{\sim}{z})_m$, not the probability itself. Write $\log \operatorname{softmax}(\underset{\sim}{z})_m$ without a fraction.

[a] https://morrison-lab.github.io/pds/expectation.html#def-cross-entropy

Solution

Solution 3.4.

$$\begin{aligned}
\log \operatorname{softmax}(\underset{\sim}{z})_m &= \log \frac{\mathrm{e}^{z_m}}{\sum_{k=1}^{C} \mathrm{e}^{z_k}} && \text{(the softmax function)} \\
&= \log \mathrm{e}^{z_m} - \log \sum_{k=1}^{C} \mathrm{e}^{z_k} && \text{(log of a quotient)} \\
&= z_m - \log \sum_{k=1}^{C} \mathrm{e}^{z_k} && \text{(log undoes exp)}
\end{aligned}$$

The first term needs no exponential at all. Only the second, the log of a sum of exponentials, can still overflow.

Definition 3.2 (Log-sum-exp)**.** The **log-sum-exp** function maps a vector $\underset{\sim}{z} \in \mathbb{R}^C$ to

$$\operatorname{LSE}(\underset{\sim}{z}) = \log \sum_{k=1}^{C} \mathrm{e}^{z_k}$$

[1, Sec. 6.2.2.3]. It is the log of the denominator of the softmax function[a]. PyTorch provides it as `torch.logsumexp`. LSE is differentiable. $\operatorname{LSE}(\underset{\sim}{z})$ is at least the largest entry of $\underset{\sim}{z}$ and at most that entry plus $\log C$. So it is called a smooth maximum.

[a] https://morrison-lab.github.io/lds/chapters/multiclass-logistic-regression.html#def-softmax

Example 3.2 (Log-sum-exp of the logits)**.**

```
lse = torch.logsumexp(logits[0], dim=0).item()
largest_logit = logits[0].max().item()
print(lse, largest_logit, largest_logit + np.log(10))
```

```
2.360166072845459 0.4042101502418518 2.7067952432358977
```

For the first test image's logits, LSE is 2.360, between the largest logit, 0.404, and that logit plus $\log 10$.

Theorem 3.1 (Log softmax). *For $\underset{\sim}{z} \in \mathbb{R}^C$ and $m = 1, \dots, C$,*

$$\log \operatorname{softmax}(\underset{\sim}{z})_m = z_m - \operatorname{LSE}(\underset{\sim}{z}).$$

Proof

Proof. By Solution 3.4 and Definition 3.2. □

Example 3.3 (Log softmax of the logits).

```
by_theorem = logits[0] - torch.logsumexp(logits[0], dim=0)
by_torch = torch.log_softmax(logits[0], dim=0)
log_softmax_gap = (by_theorem - by_torch).abs().max().item()
print(by_theorem.detach().numpy().round(3))
```

```
[-2.36  -2.384 -2.248 -2.28  -1.956 -2.32  -2.493 -2.265 -2.315 -2.518]
```

```
print(log_softmax_gap)
```

```
2.384185791015625e-07
```

For the first test image, the logits minus their log-sum-exp agree with PyTorch's `torch.log_softmax` to within 2.4e-07. Each is near $\log \frac{1}{10} \approx -2.30$.

Exercise 3.5 (Computing log-sum-exp without overflow). Let c be any number. Write $\operatorname{LSE}(\underset{\sim}{z})$ in terms of c and LSE of the shifted vector $\underset{\sim}{z} - c$, whose entries are $z_k - c$.

Solution

Solution 3.5.

$$\begin{aligned}
\operatorname{LSE}(\underset{\sim}{z}) &= \log \sum_{k=1}^{C} \mathrm{e}^{z_k} && \text{(definition of LSE)} \\
&= \log \sum_{k=1}^{C} \mathrm{e}^{c} \mathrm{e}^{z_k - c} && \text{(add and subtract } c \text{ in each exponent)} \\
&= \log \left(\mathrm{e}^{c} \sum_{k=1}^{C} \mathrm{e}^{z_k - c} \right) && \text{(take } \mathrm{e}^{c} \text{ out of the sum)} \\
&= c + \log \sum_{k=1}^{C} \mathrm{e}^{z_k - c} && \text{(log of a product)} \\
&= c + \operatorname{LSE}(\underset{\sim}{z} - c) && \text{(definition of LSE)}
\end{aligned}$$

Theorem 3.2 (Shifting log-sum-exp). *For $\underset{\sim}{z} \in \mathbb{R}^C$ and any $c \in \mathbb{R}$,*

$$\operatorname{LSE}(\underset{\sim}{z}) = c + \operatorname{LSE}(\underset{\sim}{z} - c).$$

Proof

Proof. By Solution 3.5. □

Exercise 3.6 (Choosing the shift)**.** Which c in Theorem 3.2 keeps every exponential in $\mathrm{LSE}(\underset{\sim}{z} - c)$ from overflowing?

Solution

Solution 3.6. Take $c = \max_k z_k$. Then every exponent $z_k - c$ is at most 0, so no exponential exceeds 1. One exponent is exactly 0, so the sum is at least 1, and its log is not $-\infty$.

Example 3.4 (The confident logits again)**.**

```
c = z.max()
shifted_lse = (c + torch.log(torch.exp(z - c).sum())).item()
print(shifted_lse, torch.logsumexp(z, dim=0).item())
```

```
404.21014404296875 404.21014404296875
```

```
print(torch.log_softmax(z, dim=0))
```

```
tensor([-403.7620, -428.0196, -292.4266, -324.3521,    0.0000, -364.2654,
        -537.2988, -308.9668, -359.2179, -561.8227])
```

For the logits times 1,000 of Solution 3.3, shifting by their largest entry gives LSE = 404.2, the same value `torch.logsumexp` returns, and `torch.log_softmax` gives finite log probabilities where the formula gave `nan`.

Remark (Why PyTorch's cross-entropy takes logits)

Remark 3.1 (Why PyTorch's cross-entropy takes logits). `nn.CrossEntropyLoss` and `nn.functional.cross_entropy` take logits, not probabilities. They compute the log probabilities by Theorem 3.1, with the shift of Theorem 3.2, so the loss stays finite for any logits. A classification network built for them ends with a linear layer and no softmax: softmax is applied only when probabilities are wanted for their own sake, as in Solution 3.1. Putting `nn.Softmax` at the end of the network and then calling `nn.CrossEntropyLoss` applies softmax twice.

3.3 Other kinds of output

Exercise 3.7 (An output layer for two classes)**.** An MLP is to tell 0s from 1s, predicting the probability that a digit is a 1. What should its output layer compute?

Solution

Solution 3.7. One number z, the logit of a 1, turned into a probability by the logistic function[a], $\mathrm{expit}(z)$. That is the soft 0-or-1 rule of the PyTorch chapter, with z computed by a network instead of from one feature[b].
Two outputs with softmax would also work, but they carry one number too many: $\mathrm{softmax}(z_1, z_2)_2 = \mathrm{expit}(z_2 - z_1)$, so only the difference of the two logits matters.

[a] https://morrison-lab.github.io/mds/functions.html#def-logistic-function
[b] https://morrison-lab.github.io/lds/chapters/anatomy.html#def-feature

Exercise 3.8 (A loss for two classes)**.** What loss should training minimize for the output layer of Solution 3.7?

Solution

Solution 3.8. Use the log loss[a], which is the cross-entropy[b] loss for two classes. PyTorch's `nn.BCEWithLogitsLoss` takes the logit z itself, for the reason of Remark 3.1.

[a] https://morrison-lab.github.io/lds/chapters/logistic-regression.html#def-log-loss
[b] https://morrison-lab.github.io/pds/expectation.html#def-cross-entropy

Exercise 3.9 (An output layer for a number)**.** An MLP is to predict a car's fuel economy in miles per gallon. What should its output layer compute?

Solution

Solution 3.9. One number, with no activation function: any real number can be a prediction, and squashing it with the logistic function[a] or the ReLU would rule some out.

[a] https://morrison-lab.github.io/mds/functions.html#def-logistic-function

Exercise 3.10 (A loss for a number)**.** What loss should training minimize for the output layer of Solution 3.9?

Solution

Solution 3.10. Use the mean squared error, PyTorch's `nn.MSELoss`. Minimizing the mean squared error is maximum likelihood when the output is the mean of a normal distribution with a fixed variance [1, Sec. 6.2.2.1], just as the cross-entropy[a] loss is maximum likelihood for the probabilities of the classes.

[a] https://morrison-lab.github.io/pds/expectation.html#def-cross-entropy

Remark (Output layers and their losses)

Remark 3.2 (Output layers and their losses). Solution 3.1, Solution 3.7 and Solution 3.9 pair each kind of target with an output layer and a loss, each the negative log-likelihood of a distribution for the target [1, Sec. 6.2.2]:

Target	Output layer	Distribution	PyTorch loss
A real number	One number, no activation	Normal with fixed variance	`nn.MSELoss`
One of 2 classes	One logit, logistic function[a]	Bernoulli	`nn.BCEWithLogitsLoss`
One of C classes	C logits, softmax	Categorical	`nn.CrossEntropyLoss`

4 Representational power

4.1 Training the MLP

Exercise 4.1 (Training the MLP on the digits)**.** Train the MLP of Example 2.6 on the 1,200 training images, with `nn.CrossEntropyLoss`, taking 500 steps of `torch.optim.SGD` with a learning rate of 0.5, each on the whole training set[a]. What fraction of the test images does it classify correctly?

[a] https://morrison-lab.github.io/lds/chapters/big-ideas.html#def-training-set

Solution

Solution 4.1.

```
loss_fn = nn.CrossEntropyLoss()
optimizer = torch.optim.SGD(mlp.parameters(), lr=0.5)
for step in range(500):
    optimizer.zero_grad()
    loss = loss_fn(mlp(x_train), y_train)
    loss.backward()
    optimizer.step()

with torch.no_grad():
    predicted = mlp(x_test).argmax(dim=1)
test_accuracy = (predicted == y_test).float().mean().item()
print(loss.item(), test_accuracy)
```

```
0.044963546097278595 0.9698492288589478
```

The trained network classifies 97.0% of the 597 test images correctly, against 10% for guessing. The loop is the training step of the PyTorch chapter, and the predicted digit is the one with the largest logit, which is also the one with the largest softmax probability.

4.2 Bends in a ReLU network

Exercise 4.2 (The shape of a small ReLU network)**.** An MLP has one input $z \in \mathbb{R}$, one hidden layer of L ReLU units, and one output with no activation function:

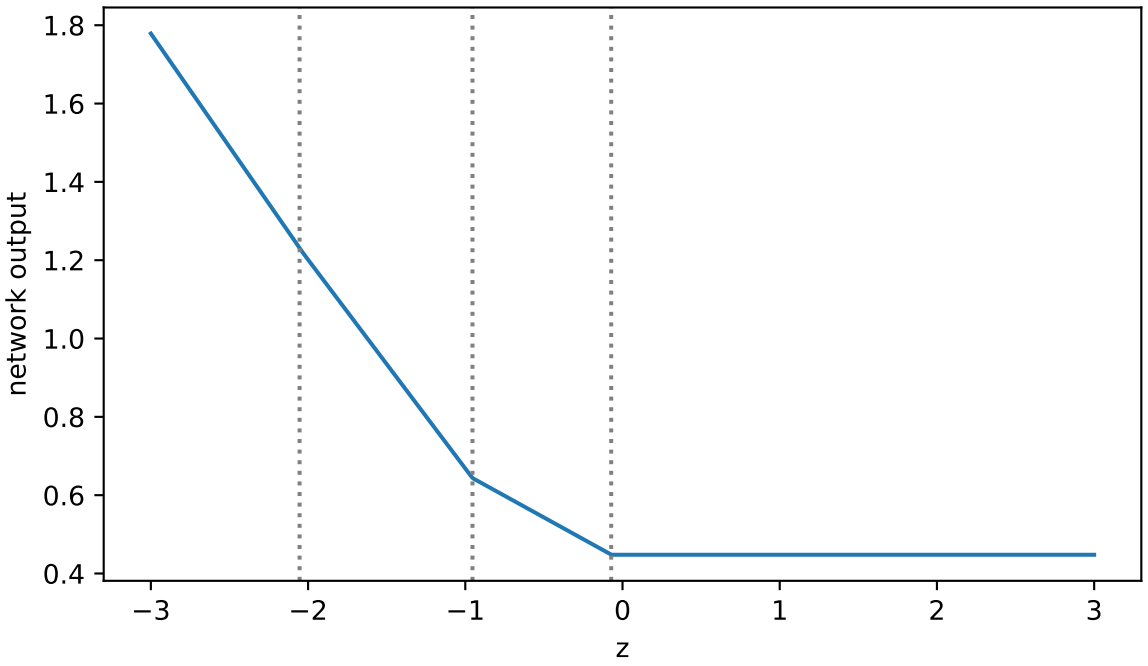
$$\hat{y} = c + \sum_{i=1}^{L} v_i \max(0, w_i z + b_i).$$

What does the graph of $\hat{y}$ against z look like?

Solution

Solution 4.2. For $w_i \neq 0$, unit i is 0 on one side of $z = -b_i/w_i$ and the line $w_i z + b_i$ on the other, so its graph is two straight pieces that meet at that point. A weighted sum of such graphs, plus a constant, is still made of straight pieces. It can change slope only where some unit does. So the graph of $\hat{y}$ is a connected chain of straight pieces, with at most L bends.

```
_ = torch.manual_seed(3)
tiny = nn.Sequential(nn.Linear(1, 3), nn.ReLU(), nn.Linear(3, 1))
z = torch.linspace(-3, 3, 601).unsqueeze(1)
with torch.no_grad():
    y_hat = tiny(z)
    bends = (-tiny[0].bias / tiny[0].weight.squeeze()).numpy()
fig, ax = plt.subplots(figsize=(6, 3.5), layout="constrained")
ax.plot(z, y_hat);
for bend in bends:
    ax.axvline(bend, color="gray", linestyle=":");
ax.set_xlabel("z");
ax.set_ylabel("network output");
plt.show()
```



For a random network with $L = 3$, the dotted lines mark the three points $-b_i/w_i$, and the graph bends at each one that lies in the plotted range.

Exercise 4.3 (Following any curve)**.** Let f be a continuous function on an interval $[a, b]$. Could a network like the one in Exercise 4.2 follow f as closely as you like?

Solution

Solution 4.3. Yes, with enough units. Mark points on the graph of f at closely spaced values of z, and join neighbouring points with straight lines. Because f is continuous on a closed interval, it changes little between nearby points, so the joined-up graph stays as close to f as you like once the points are close enough together. Call the inner points $t_1 < \cdots < t_{n-1}$, and the slopes of the n pieces $s_1, \ldots, s_n$. A network like the one in Exercise 4.2 draws the joined-up graph on $[a, b]$ with n units, one per piece, each with $w_i = 1$:

- $c = f(a)$ starts the graph at the first point;
- the unit $s_1 \max(0, z - a)$ gives the first piece its slope;
- for each inner point t_j, the unit $(s_{j+1} - s_j)\max(0, z - t_j)$ is 0 to the left of t_j, and changes the slope from s_j to s_{j+1} at t_j.

Theorem 4.1 (Universal approximation)**.** *Let g be a continuous activation function that is not a polynomial, such as the logistic function[a] or the ReLU. For every continuous function f on a closed and bounded set $\mathcal{X} \subset \mathbb{R}^p$ and every $\epsilon > 0$, there is a width[b] L and parameters[c] $\underset{\sim}{w}_i \in \mathbb{R}^p$ and $b_i, v_i, c \in \mathbb{R}$ such that*

$$\left| f(\underset{\sim}{x}) - \left(c + \sum_{i=1}^{L} v_i \, g(\underset{\sim}{w}_i^\top \underset{\sim}{x} + b_i) \right) \right| < \epsilon \quad \text{for every } \underset{\sim}{x} \in \mathcal{X}.$$

[1, Sec. 6.4.1], [5].

[a] https://morrison-lab.github.io/mds/functions.html#def-logistic-function
[b] https://morrison-lab.github.io/lds/chapters/neural-nets.html#def-width
[c] https://morrison-lab.github.io/lds/chapters/big-ideas.html#def-parameter

Proof

Proof. For one input and the ReLU, Solution 4.3 gives the idea. Cybenko [6] proved the theorem for logistic-shaped activation functions, Hornik [7] for bounded nonconstant ones, and Leshno et al. [5] for every activation function that is not a polynomial; those proofs need more analysis than these notes assume. □

Example 4.1 (Universal approximation for one input)**.** For one input, the ReLU and a closed interval $[a, b]$, Solution 4.3 builds the network that Theorem 4.1 promises: it joins points on the graph of f with straight pieces, and gives the network one unit for each piece. Making the pieces shorter brings the network closer to f, at the cost of more units.

Exercise 4.4 (Wider networks on the Auto data)**.** Fit networks with one hidden layer of ReLU units, of width[a] 1, 3 and 20, to miles per gallon on horsepower in the Auto data[b]. How does the training mean squared error change as the width grows?

[a] https://morrison-lab.github.io/lds/chapters/neural-nets.html#def-width
[b] https://morrison-lab.github.io/lds/chapters/course-overview.html#def-auto-data

Solution

Solution 4.4.

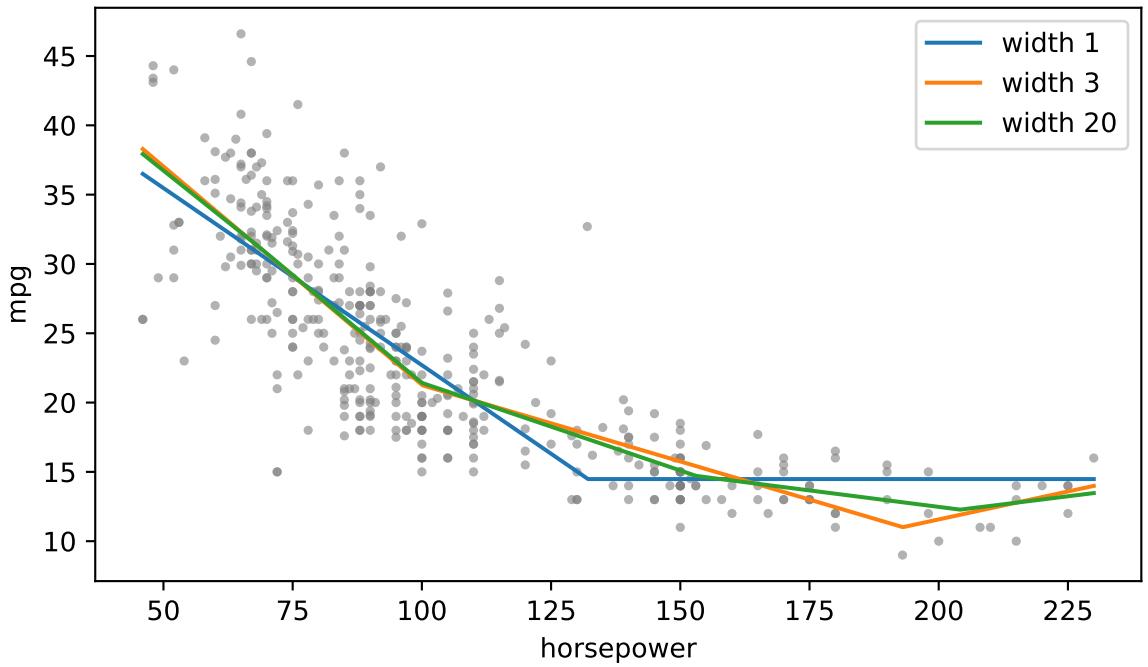
```
auto = pd.read_csv("data/auto.csv")
hp = torch.tensor(auto["horsepower"].to_numpy(), dtype=torch.float32).unsqueeze(1)
mpg = torch.tensor(auto["mpg"].to_numpy(), dtype=torch.float32).unsqueeze(1)
hp_scaled = (hp - hp.mean()) / hp.std()
print(auto[["horsepower", "mpg"]].head(3))
```

```
   horsepower   mpg
0         130  18.0
1         165  15.0
2         150  18.0
```

```
n_steps = 3000


def fit_width(width, steps=n_steps):
    # Fit a one-hidden-layer ReLU network of `width` units to mpg on horsepower.
    torch.manual_seed(2)
    net = nn.Sequential(nn.Linear(1, width), nn.ReLU(), nn.Linear(width, 1))
    optimizer = torch.optim.SGD(net.parameters(), lr=0.01)
    for step in range(steps):
        optimizer.zero_grad()
        loss = nn.functional.mse_loss(net(hp_scaled), mpg)
        loss.backward()
        optimizer.step()
    return net, loss.item()


grid = torch.linspace(hp.min().item(), hp.max().item(), 300).unsqueeze(1)
fig, ax = plt.subplots(figsize=(6, 3.5), layout="constrained")
ax.scatter(hp, mpg, s=6, color="gray", alpha=0.6);
auto_mse = {}
for width in [1, 3, 20]:
    net, auto_mse[width] = fit_width(width)
    with torch.no_grad():
        ax.plot(grid, net((grid - hp.mean()) / hp.std()), label=f"width {width}");
ax.set_xlabel("horsepower");
ax.set_ylabel("mpg");
ax.legend();
plt.show()
```




```
print(auto_mse)
```

```
{1: 19.619924545288086, 3: 18.540300369262695, 20: 18.385217666625977}
```

The Auto data[a] hold 392 cars from the UCI Auto MPG dataset [8], from a local copy, `data/auto.csv`[b], of the version in lds. Each network fits miles per gallon on horsepower, rescaled to mean 0 and standard

deviation 1, by 3,000 steps of gradient descent on the mean squared error. With 1 unit the fit can bend only once, and its training mean squared error is 19.6. With 3 units it is 18.5, and with 20 it is 18.4. So the error falls as the network gets wider. Theorem 4.1 says that a wide enough network can follow any continuous curve; it does not say that following the training data[c] more closely predicts new cars better.

[a] https://morrison-lab.github.io/lds/chapters/course-overview.html#def-auto-data
[b] https://morrison-lab.github.io/dds/chapters/data/auto.csv
[c] https://morrison-lab.github.io/lds/chapters/anatomy.html#def-training-data

ℹ Remark (What universal approximation leaves out)

Remark 4.1 (What universal approximation leaves out). Theorem 4.1 says that some network of one hidden layer comes within ϵ of f. It does not say how wide that network must be, a width[a] that can grow very fast with the number of inputs p, nor that training finds its parameters[b], nor that it predicts new data well [1, Sec. 6.4.1].
Depth[c] helps with the first of these. A ReLU network's graph is made of straight pieces, as Solution 4.2 showed for one input. For networks with the same total number of units, the number of pieces a deep network can make grows exponentially with its number of layers, while a network with one hidden layer makes far fewer [9]. Some functions with many pieces therefore need far fewer units in a network with several layers.

[a] https://morrison-lab.github.io/lds/chapters/neural-nets.html#def-width
[b] https://morrison-lab.github.io/lds/chapters/big-ideas.html#def-parameter
[c] https://morrison-lab.github.io/lds/chapters/neural-nets.html#def-depth

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[9] https://doi.org/10.1037/h0042519
[10] https://doi.org/10.1016/S0893-6080(05)80131-5
[11] https://doi.org/10.1007/BF02551274
[12] https://doi.org/10.1016/0893-6080(91)90009-T
[13] https://doi.org/10.24432/C5859H